Assignment 2

CSED441 – Introduction to Computer Vision

TA: Jaehyun Pahk (jhpahk@postech.ac.kr)

Due: November 2nd (Sunday), 23:59 KST

Overview

This assignment consists of two parts with the following distribution:

1. **Code Implementation**: 70 points
2. **Discussion and Report**: 30 points

The goal of this assignment is to gain hands-on experience with:

- **Histogram of Oriented Gradients (HOG)** feature descriptor
- **Support Vector Machine (SVM)** classifier
- **Sliding window object detector**

You will implement key functions and then conduct parameter experiments to observe how different hyperparameters influence detection performance. You can complete the assignment by following the steps in `assignment2.ipynb` in order.

Dataset

You will need to download the **INRIA Person Dataset** to complete this assignment. This dataset contains images of pedestrians and non-pedestrians for training and testing your implementation.

Link: https://drive.google.com/file/d/1wTxod2BhY_HUkEdDYRVSuw-nDuqrgCu7/view

Submission Format

Report submission:

Name your report as a single PDF file named: **`assign2_[your_student_ID].pdf`**

Your report must include the following things:

- Experiment with different SVM hyperparameters and report their impact on classification performance.

- Experiment with sliding window detection parameters and analyze the balance between computational efficiency and detection quality.

- Your SVM classifier likely achieves decent performance on cropped patches, but the sliding window detector's performance on full images may be disappointing. Analyze this gap and propose potential approaches to improve detection performance.

Note: **You don't need to perform exhaustive hyperparameter tuning** – focus on exploring a few key parameters and understanding their effects.

Code submission:

- You will be provided with a starter code package (`release.zip`).
- Unzip the file, and rename the top-level directory to your student ID.
- Implement and modify the code inside this directory. Ensure that all required functions are completed.
- **You must add clear comments in your code**.
- After completing this assignment, re-compress the directory back into a zip file named: `assign2_[your_student_ID].zip`
- Submit this zip file together with your report.

If you have any questions about the assignment, please feel free to reach out to the TA!